

Enhancing Risk Management Effectiveness in Hybrid Projects through AI-Driven Strategic Governance

A Structural Equation Modelling Study

Based on primary data from 384 project professionals

Methodology: Structural Equation Modelling (SEM) | SPSS & AMOS

Presentation Outline

0
1

Introduction & Background

0
2

Research Objectives & Questions

0
3

Conceptual Framework & Hypotheses

0
4

Methodology

0
5

Demographics & Sample Profile

0
6

Measurement Model & Validity

0
7

SEM Results (H1 – H5)

0
8

Discussion & Implications

0
9

Conclusion & Future Research

Introduction & Background

The Problem

Traditional risk management relies on static registers, periodic assessments, and subjective expert judgment.

Hybrid Project Complexity

Agile + Waterfall integration creates fragmented governance, inconsistent risk documentation, and ambiguous escalation.

The AI Opportunity

AI enables predictive analytics, real-time monitoring, and data-driven decision support for risk governance.

Most IT projects fail to meet original objectives
(Standish Group, 2020)

Traditional methods struggle with dynamic,
data-intensive environments

AI: Machine Learning + NLP + Deep Learning
enable continuous risk sensing

Research Objectives & Questions

R
O1

Examine the impact of Risk Process Alignment on Risk Management Effectiveness

R
O2

Analyze the influence of Risk Visibility Integration on Risk Management Effectiveness

R
O3

Assess the effect of Timely Risk Escalation on Risk Management Effectiveness

R
O4

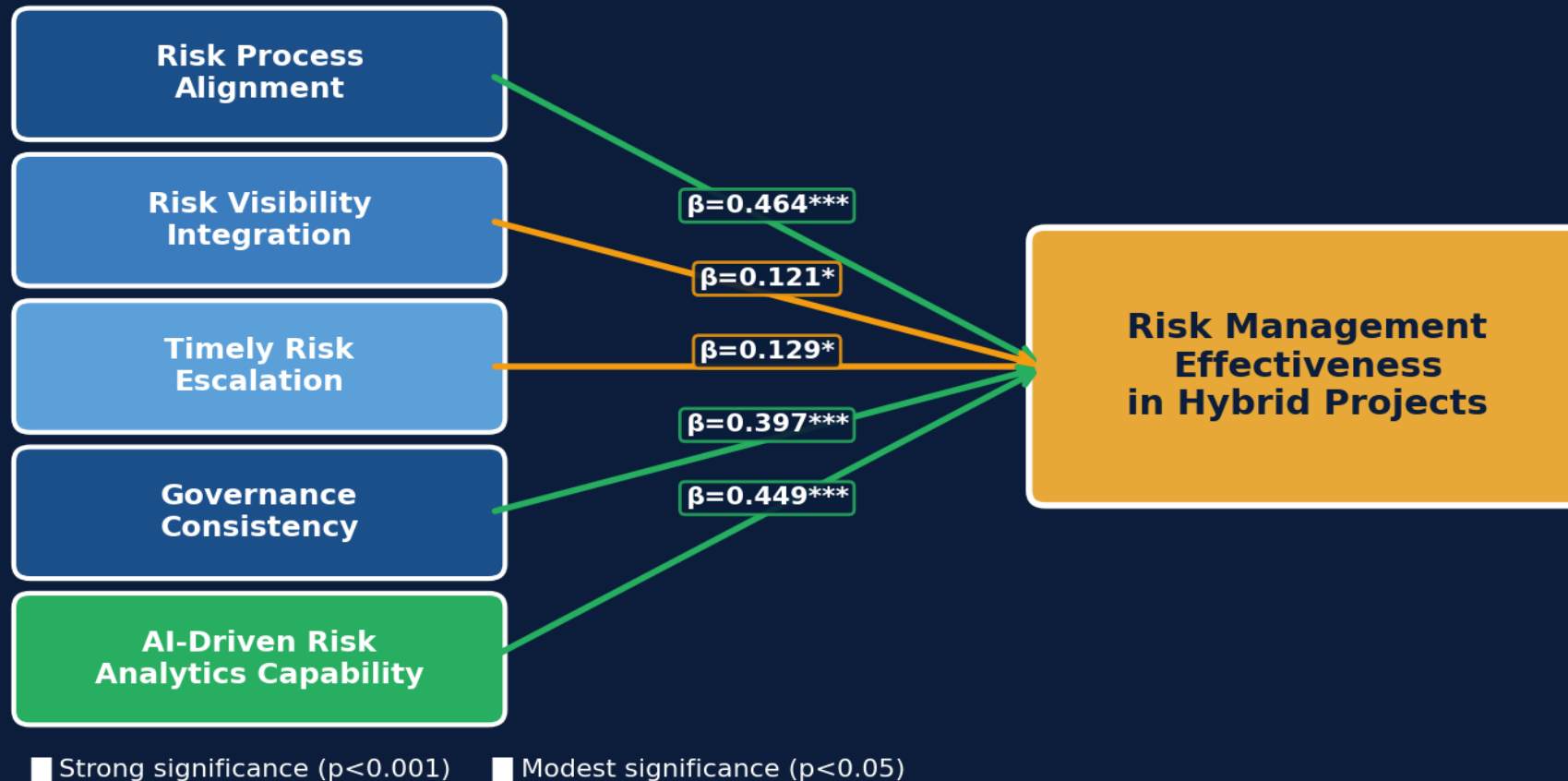
Evaluate the role of Governance Consistency in enhancing Risk Management Effectiveness

R
O5

Examine the impact of AI-Driven Risk Analytics Capability on Risk Management Effectiveness

Conceptual Framework

Integrated Model of AI-Driven Strategic Risk Management



Research Hypotheses

H1

Risk Process Alignment has a significant positive impact on Risk Management Effectiveness in Hybrid Project Environments.

H2

Risk Visibility Integration has a significant positive impact on Risk Management Effectiveness in Hybrid Project Environments.

H3

Timely Risk Escalation has a significant positive impact on Risk Management Effectiveness in Hybrid Project Environments.

H4

Governance Consistency has a significant positive impact on Risk Management Effectiveness in Hybrid Project Environments.

H5

AI-Driven Risk Analytics Capability has a significant positive impact on Risk Management Effectiveness in Hybrid Project Environments.

Research Methodology

Research Design

Quantitative,
cross-sectional
survey design

Sample Size

384 project
professionals
across industries

Data Collection

Structured
questionnaire
(5-point Likert)

Analysis Tools

SPSS + AMOS
for CFA and
SEM analysis

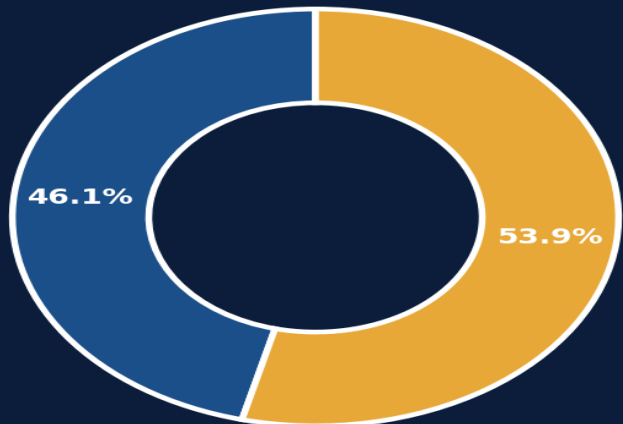
Statistical Techniques Applied

- Confirmatory Factor Analysis (CFA) for measurement model validation
- Structural Equation Modelling (SEM) for hypothesis testing
- Cronbach's Alpha, AVE, and Composite Reliability for construct validity
- KMO and Bartlett's Test for sampling adequacy (KMO = 0.927)
- Discriminant Validity via Fornell-Larcker Criterion

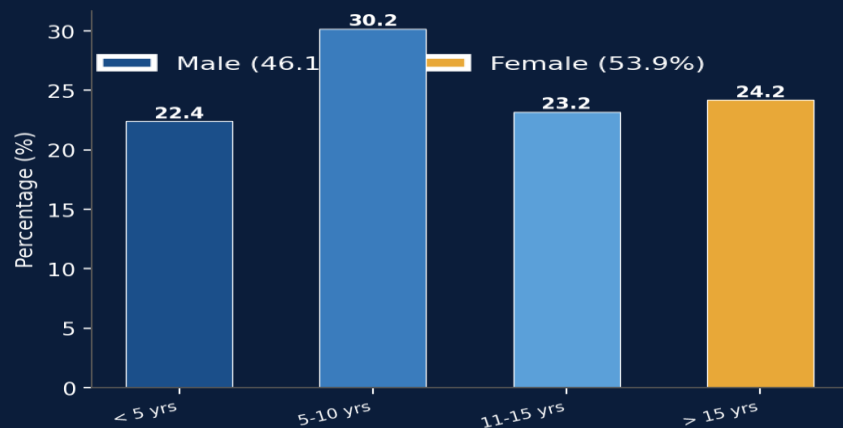
Sample Demographics

N = 384 Project Professionals

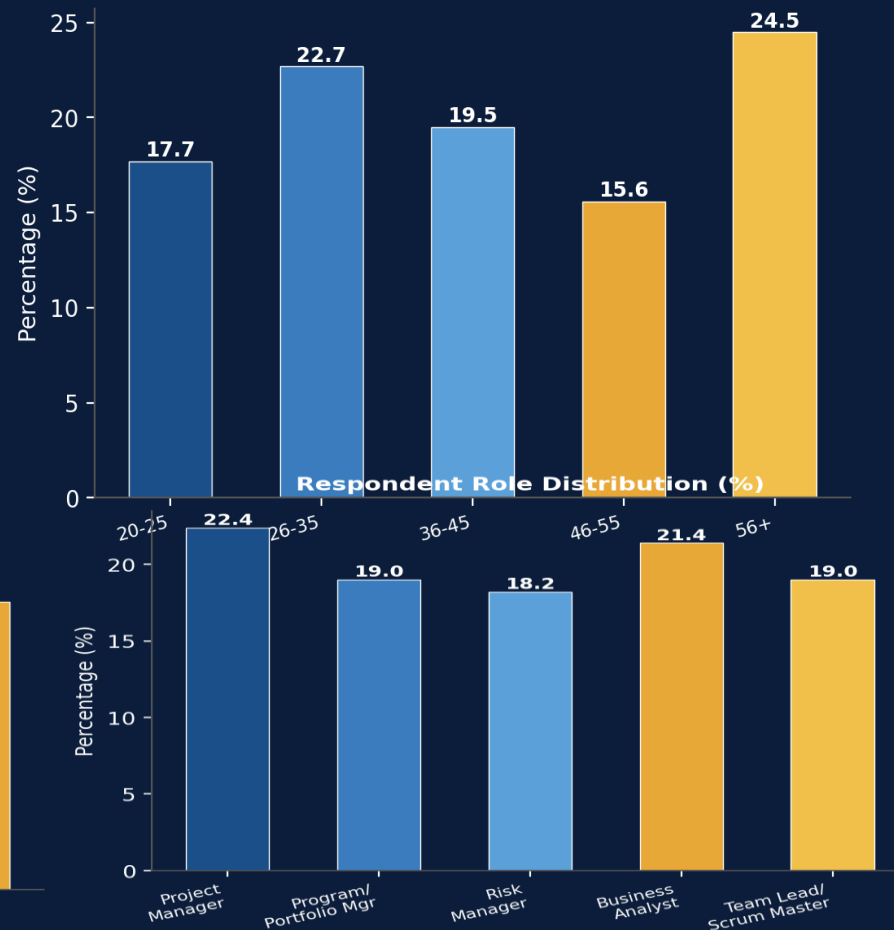
Gender Distribution



Project Management Experience (%)



Age Distribution (%)



77%+ have over 5 years experience

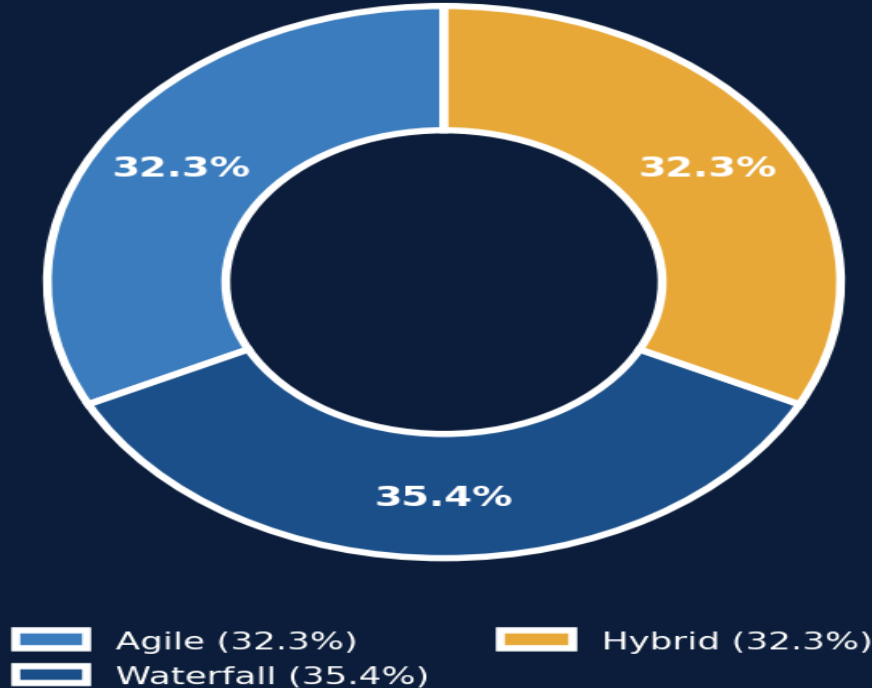
53.9% Female
46.1% Male

27.6% Hold Professional Certifications

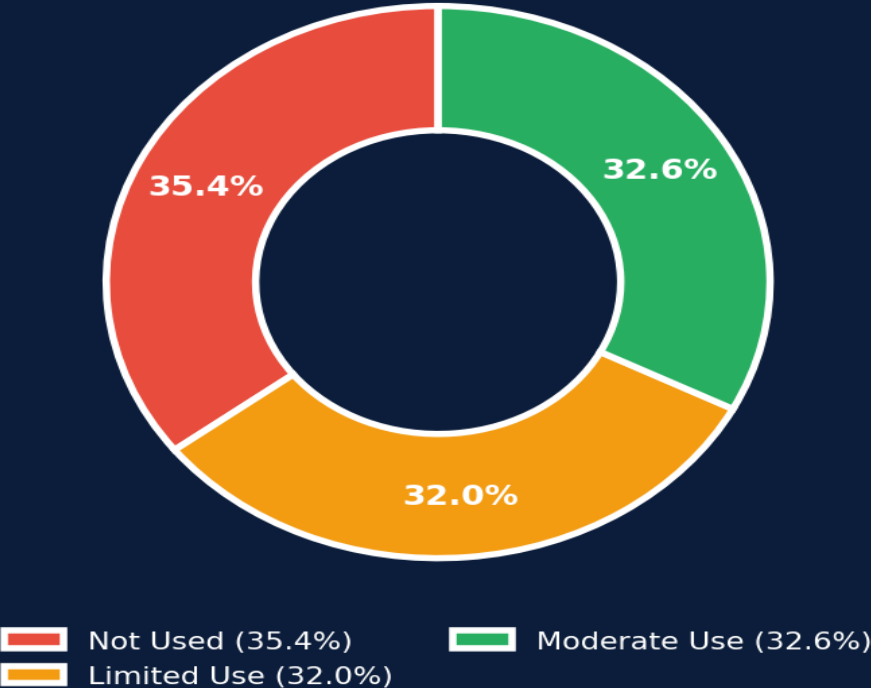
Balanced role distribution

Methodology & AI Adoption Profile

Primary Project Methodology



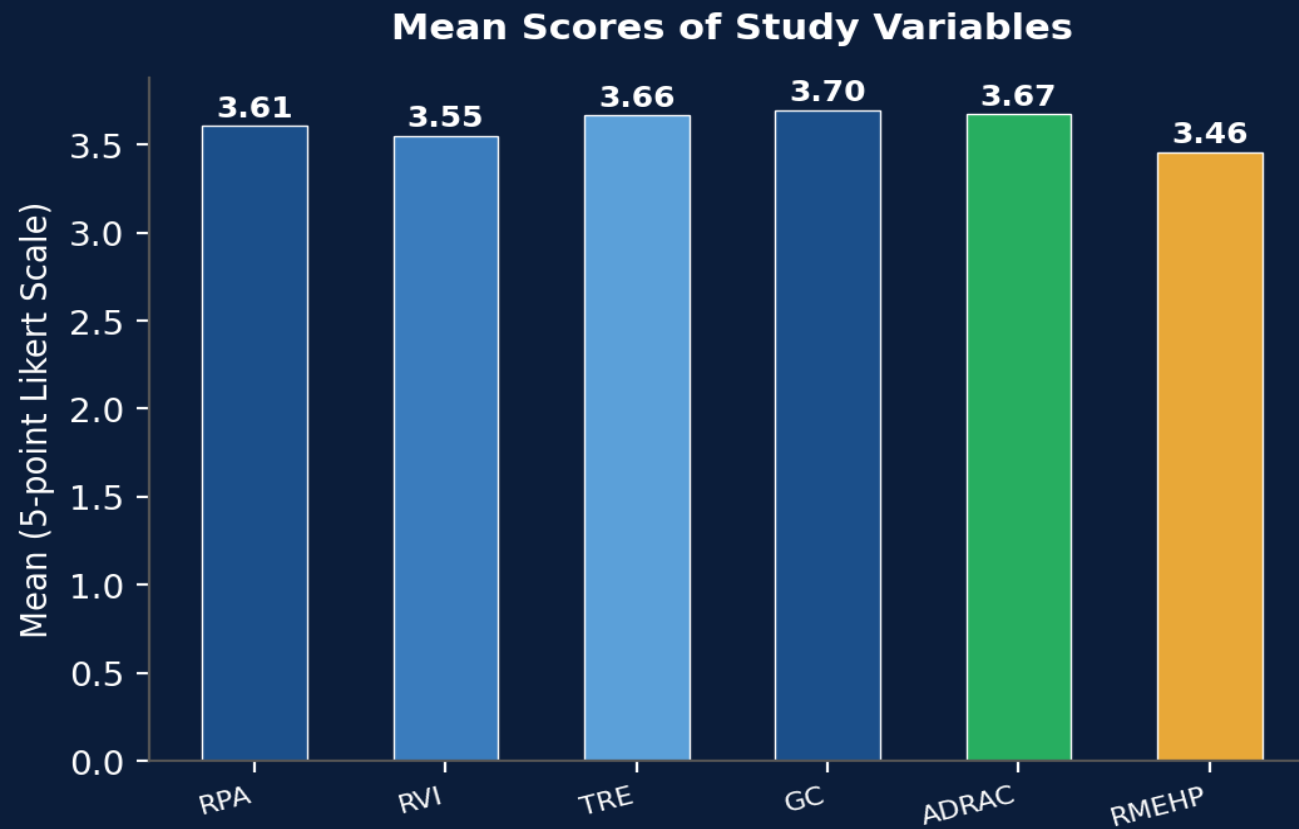
Extent of AI Usage in Risk Management



Even distribution across methodologies ensures balanced insights | AI adoption is progressing but not yet universal

Descriptive Statistics

Mean Scores on 5-point Likert Scale



Key Observations

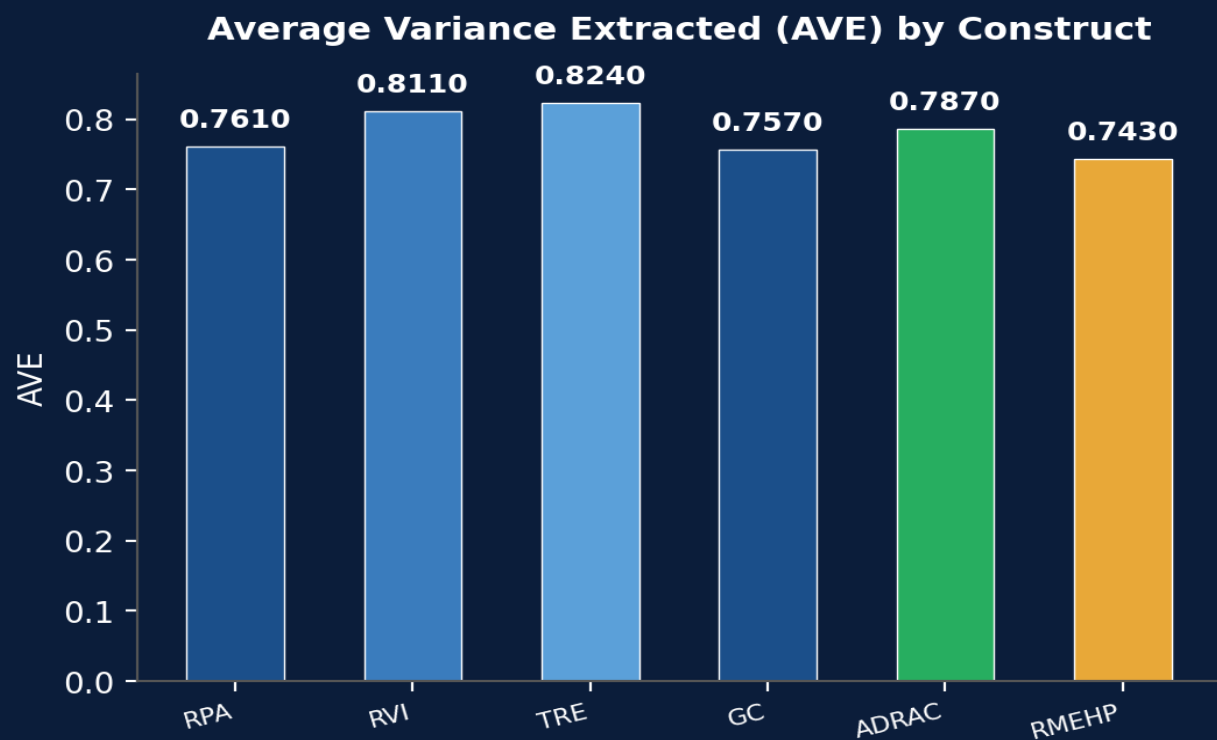
Governance Consistency and AI-Driven Risk Analytics score highest (3.70, 3.67)

Risk Management Effectiveness has the lowest mean (3.46), suggesting uneven translation of practices into outcomes

All skewness values within ± 2 and kurtosis within $\pm 3 \rightarrow$ data is suitable for SEM

All constructs show moderately high agreement

Measurement Model & Validity



All AVE > 0.5 and CR > 0.7 confirm convergent validity

Convergent Validity

Construct	AVE	CR
Risk Process Alignment	0.761	0.858
Risk Visibility Integration	0.811	0.873
Timely Risk Escalation	0.824	0.877
Governance Consistency	0.757	0.880
AI-Driven Risk Analytics	0.787	0.866
Risk Mgmt Effectiveness	0.743	0.852

KMO = 0.927 (Excellent Sampling Adequacy)
Bartlett's Test: $\chi^2 = 7618.445$, $p < 0.001$

Hypothesis 1 – Result

Risk Process Alignment → Risk Management Effectiveness

Hypothesis 1



C.R. = 7.346 | $p = <0.001$

H1 Supported

Result: H1 Supported

$\beta = 0.464$ | C.R. = 7.346 | $p < 0.001$

Model Fit:

CMIN/DF = 2.366 | GFI = 0.958

CFI = 0.976 | RMSEA = 0.060

Interpretation:

Risk Process Alignment shows a moderate and meaningful positive relationship with RME.

When risk processes are aligned across Agile & Waterfall, organizations experience improved coordination and reduced ambiguity.

Hypothesis 2 – Result

Risk Visibility Integration → Risk Management Effectiveness

Hypothesis 2



C.R. = 2.116 | $p = 0.034$

H2 Supported (Marginal)

Result: H2 Supported (Marginal Effect)

$\beta = 0.121$ | C.R. = 2.116 | $p = 0.034$

Model Fit:

CMIN/DF = 1.376 | CFI = 0.994

RMSEA = 0.031

Interpretation:

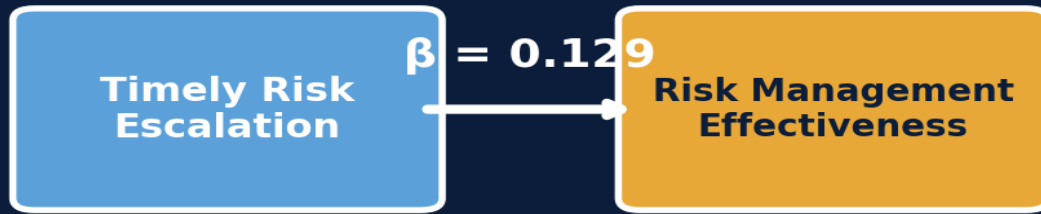
While statistically significant, the effect size is modest.

Risk visibility alone is insufficient — it functions as a supportive mechanism rather than a dominant driver when governance structures are weak.

Hypothesis 3 – Result

Timely Risk Escalation → Risk Management Effectiveness

Hypothesis 3



C.R. = 2.283 | $p = 0.022$

H3 Supported (Modest)

Result: H3 Supported (Modest Effect)

$\beta = 0.129$ | C.R. = 2.283 | $p = 0.022$

Model Fit:

CMIN/DF = 1.326 | CFI = 0.995

RMSEA = 0.029

Interpretation:

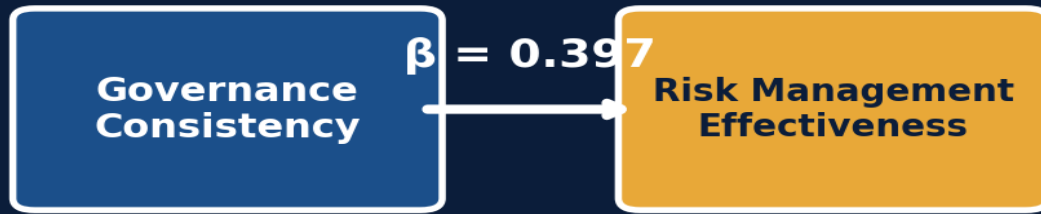
Statistically significant but modest direct effect.

Agile teams manage risks locally, reducing the need for frequent formal escalation. Effectiveness depends on appropriateness, not speed.

Hypothesis 4 – Result

Governance Consistency → Risk Management Effectiveness

Hypothesis 4



C.R. = 6.869 | $p = <0.001$

H4 Supported

Result: H4 Supported

$\beta = 0.397$ | C.R. = 6.869 | $p < 0.001$

Model Fit:

CMIN/DF = 1.716 | CFI = 0.985

RMSEA = 0.043

Interpretation:

Governance Consistency is a structural backbone that stabilizes risk management across diverse execution approaches.

Uniform governance reduces ambiguity in authority, accountability, and decision-making in hybrid settings.

Hypothesis 5 – Result

AI-Driven Risk Analytics Capability → Risk Management Effectiveness

Hypothesis 5



C.R. = 7.697 | $p = <0.001$

H5 Supported

Result: H5 Supported

$\beta = 0.449$ | C.R. = 7.697 | $p < 0.001$

Model Fit:

CMIN/DF = 1.995 | CFI = 0.982

RMSEA = 0.051

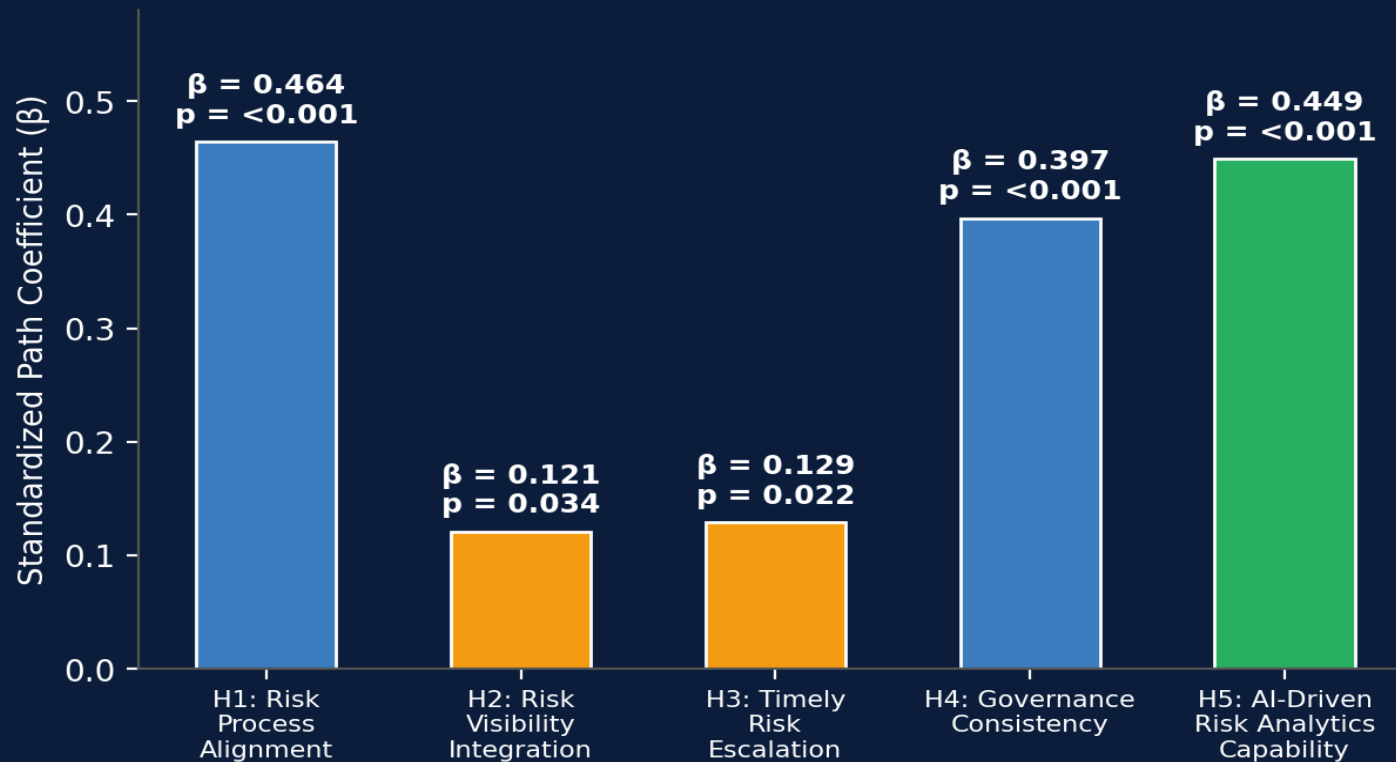
Interpretation:

AI-Driven Risk Analytics is one of the STRONGEST predictors of RME.

AI enhances predictive foresight, anomaly detection, real-time monitoring, and automated risk prioritization in hybrid environments.

Summary of Hypothesis Testing Results

SEM Path Coefficients - All Hypotheses



Hypothesis Outcomes

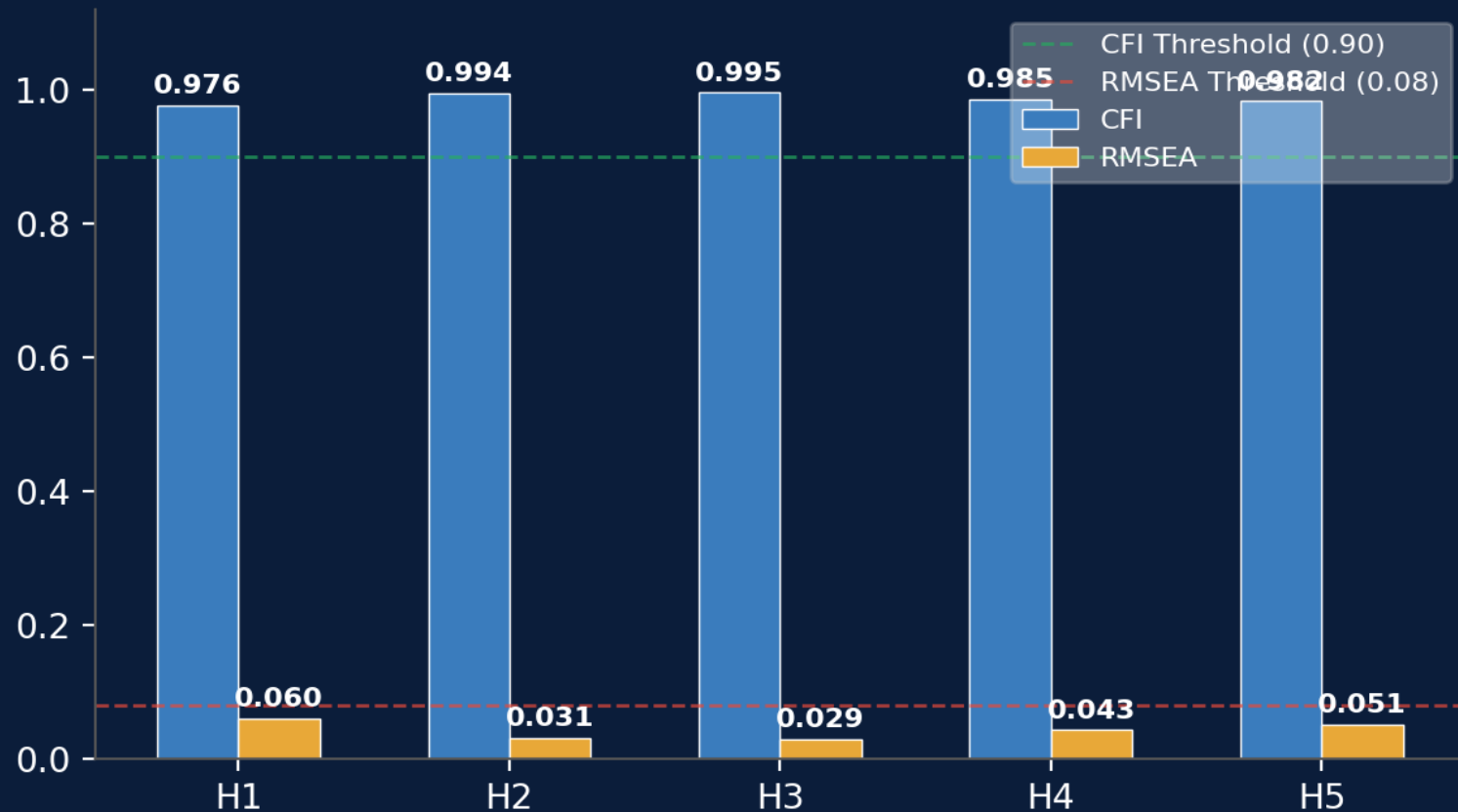
H1	RPA → RME	$\beta=0.464$	Supported
H2	RVI → RME	$\beta=0.121$	Marginal
H3	TRE → RME	$\beta=0.129$	Modest
H4	GC → RME	$\beta=0.397$	Supported
H5	ADRAC → RME	$\beta=0.449$	Supported

Strongest: H5 (AI) > H1 (RPA) > H4 (GC)

Model Fit Indices

All models demonstrate good-to-excellent fit

Model Fit Indices Across Hypotheses



Acceptance Criteria

- ✓ CMIN/DF < 3.00
- ✓ GFI ≥ 0.90
- ✓ NFI ≥ 0.90
- ✓ CFI ≥ 0.90
- ✓ IFI ≥ 0.90
- ✓ RMSEA ≤ 0.08
- ✓ RMR ≤ 0.05

All models meet or exceed all thresholds

Discussion & Key Insights

Structural Coherence Matters Most

Governance Consistency and Risk Process Alignment are foundational. Effective risk management requires deliberate integration across Agile and Waterfall governance systems.

AI as Strategic Capability

AI analytics is not merely a technical tool but an organizational capability that enhances predictive foresight and decision quality. It complements and amplifies governance structures.

Visibility & Speed Are Insufficient Alone

Risk visibility and timely escalation are supportive but not dominant drivers. Without strong governance and interpretive capacity, information availability does not improve outcomes.

Theoretical & Practical Implications

Theoretical Contributions

- Extends project risk management theory by positioning AI as a structural determinant, not just a tool
- Advances hybrid project management research through SEM-based empirical validation
- Bridges governance theory and dynamic capability theory in hybrid project domain
- Demonstrates AI capability as complementary to governance, not a replacement

Practical Recommendations

- Prioritize harmonizing risk processes across Agile and Waterfall components
- Establish consistent governance standards: accountability, decision rights, and escalation pathways
- Invest in AI analytics embedded within governance frameworks, not standalone
- Train managers to interpret and act on AI-generated insights for strategic decisions

Conclusion

Effective risk management in hybrid projects requires integrated structural alignment, governance stability, and AI-driven analytical intelligence — not isolated practices.

1

Governance + Process Alignment

Structural coherence is the foundation — consistent governance and aligned risk processes drive effectiveness.

2

AI as Strategic Enabler

AI capability is one of the strongest predictors — it enhances predictive foresight and amplifies governance.

3

Beyond Visibility & Speed

Organizations must focus on decision quality and governance clarity, not just information availability.

Limitations & Future Research

Limitations

- Cross-sectional design limits causal inference
- Sample drawn from India may limit generalizability
- Self-reported perceptions may introduce response bias
- Focuses on direct relationships only (no mediation/moderation)

Future Research Directions

- Longitudinal studies across project phases
- Cross-cultural replication in different regions
- Explore mediating roles of organizational culture and project complexity
- Examine AI interaction with specific Agile practices (sprints, retrospectives)

Thank You

Questions & Discussion Welcome